

Empirical Research in Finance

8. Fama–French 3-factor model

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Today builds on...

- ▶ Class 03 – interpreting regression output (α , β , R^2 , t-stats)
- ▶ Class 05 – time-series and panel regressions
- ▶ Today — a regression model with multiple risk factors as regressors

Why Do We Need Asset Pricing Models?

- ▶ Investors want to understand **why returns differ**
- ▶ Is higher return just **luck**, or **risk**?
- ▶ Asset pricing models link **risk** to **expected return**

The Starting Point: CAPM

- ▶ Capital Asset Pricing Model (CAPM)
- ▶ Says only **market risk** matters

$$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$$

Problem

CAPM does not explain returns very well in practice

What did Fama and French notice?

- ▶ Some stocks earn higher returns than predicted by CAPM
- ▶ Small firms often outperform large firms – Size Premium
- ▶ Value stocks often outperform growth stocks – Value Premium
 - ▶ Value stocks: high book-to-market ratios
 - ▶ Growth stocks: low book-to-market ratios

What is book-to-market?

$$B/M = \frac{\text{Book equity (from the balance sheet)}}{\text{Market equity (price} \times \text{shares)}}$$

Where book equity $\approx$ Assets – Liabilities.

Firm	Book equity	Market cap	B/M
A (“value”)	€100m	€50m	2.00
B (“neutral”)	€100m	€100m	1.00
C (“growth”)	€50m	€500m	0.10

A high B/M means the market values the firm at *less* than its accounting book value — often a mature firm, with tangible assets, in a slow industry.

The Big Idea

Fama–French (1993)

Stock returns are explained by **multiple risk factors**, not just the market.

- ▶ Market risk
- ▶ Size risk
- ▶ Value risk

Key: FF factors explain return patterns statistically, but not the underlying economic mechanism.

What is a factor, exactly?

Definition

A factor is a *portfolio* that captures a specific source of risk.

- ▶ Built from many stocks
- ▶ Long some stocks, short others
- ▶ Typically rebalanced once per year

General factor construction logic

1. Choose a firm characteristic (size, value, ...)
2. Sort stocks by that characteristic
3. Form portfolios
4. Take return differences

This same recipe is used to build every factor you will meet today — and most of the factors in the modern literature.

The three Fama–French factors — names and intuition

1. **Market:** $R_m - R_f$
same as CAPM — compensation for overall market risk
2. **SMB** (Small Minus Big): small-cap return — large-cap return
compensation for size risk
3. **HML** (High Minus Low): high B/M return — low B/M return
compensation for value risk

Writing it as a regression

For a portfolio p :

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{m,p}(R_{m,t} - R_{f,t}) + \beta_{S,p}R_{\text{SMB},t} + \beta_{H,p}R_{\text{HML},t} + \varepsilon_{p,t}$$

In plain words:

- ▶ LHS: the portfolio's excess return
- ▶ β_m : how much the portfolio moves with the market
- ▶ β_S : how much it loads on small firms
- ▶ β_H : how much it loads on value firms
- ▶ α : the part of the return the three factors cannot explain

Factor 1: Market Risk

- ▶ Same as CAPM
- ▶ Excess return of the market

$$R_M - R_f$$

Intuition

Investors require compensation for overall market risk.

Factor 2: Size (SMB)

SMB = Small Minus Big

- ▶ Measure market capitalisation
- ▶ Split firms into
 - ▶ Small (bottom 50%)
 - ▶ Big (top 50%)
- ▶ $\text{SMB} = \text{return on small portfolio} - \text{return on big portfolio}$

Worked example, part 1: simple SMB

10 stocks. Returns this month:

	Small (5 stocks)	Big (5 stocks)
Growth (L)	1 stock: -10%	3 stocks: -10% each
Medium (M)	1 stock: 0%	1 stock: 0%
Value (H)	3 stocks: +10% each	1 stock: +10%

Naive (simple) SMB just averages all small minus all big:

$$R_{\text{Small}} = \frac{-10+0+10+10+10}{5} = +4\% \quad R_{\text{Big}} = \frac{-10-10-10+0+10}{5} = -4\%$$

$$\text{Simple SMB} = +4\% - (-4\%) = +8\%$$

Worked example, part 2: double-sorted SMB

Compute each cell's portfolio return first, then average:

$$\begin{aligned}\text{SMB}_{\text{FF}} &= \underbrace{\frac{1}{3}(R_{SL} + R_{SM} + R_{SH})}_{\text{small side, equal weight on L/M/H}} - \underbrace{\frac{1}{3}(R_{BL} + R_{BM} + R_{BH})}_{\text{big side, equal weight on L/M/H}} \\ &= \frac{-10+0+10}{3} - \frac{-10+0+10}{3} = 0\% - 0\% = \boxed{0\%}\end{aligned}$$

Why so different? Simple SMB weights by how many stocks fall into each bucket, and in our data:

- ▶ Small stocks are mostly value (3/5)
- ▶ Big stocks are mostly growth (3/5)

Double sorting forces equal value composition on both sides — and only then does pure size show through.

Factor 3: Value (HML)

HML = High Minus Low

- ▶ Compute book-to-market ratios of equity
- ▶ Split firms into
 - ▶ Low (growth) — bottom 30%
 - ▶ Medium — middle 40%
 - ▶ High (value) — top 30%
- ▶ HML = return on high portfolio — return on low portfolio

Double sorting: the 2×3 grid

		Book-to-market →		
		Low (growth)	Medium	High (value)
Size ↓	Big	BL	BM	BH
	Small	SL	SM	SH

SMB averages the three small cells, minus the three big cells (rows).

HML averages the two value cells, minus the two growth cells (columns).

Double sorting: the formulas

$$\text{SMB} = \frac{1}{3}(R_{SL} + R_{SM} + R_{SH}) - \frac{1}{3}(R_{BL} + R_{BM} + R_{BH})$$

$$\text{HML} = \frac{1}{2}(R_{SH} + R_{BH}) - \frac{1}{2}(R_{SL} + R_{BL})$$

SMB is *value-neutral* (equal L/M/H on each side).

HML is *size-neutral* (equal S/B on each side).

What do the betas mean?

Output	In plain words
$\beta_m = 1.0$	Excess returns move one-for-one with the market.
$\beta_S = +0.8$	Tilts toward small-cap stocks.
$\beta_S = -0.5$	Tilts toward large-cap / big-firm stocks.
$\beta_H = +0.6$	Tilts toward value stocks.
$\beta_H = -0.4$	Tilts toward growth stocks.

Key point

Different portfolios load differently on these risks. A US tech ETF and a European bank ETF will likely have very different $(\beta_m, \beta_S, \beta_H)$.

Interpretation of alpha

Alpha

The average return *not* explained by exposure to the three factors.

- ▶ $\alpha \approx 0$: model explains returns well
- ▶ $\alpha > 0$ (and significant): outperformance after risk adjustment
- ▶ $\alpha < 0$: underperformance after risk adjustment

Empirical performance: does it work?

- ▶ On 25 size/B-M sorted portfolios, the average $|\alpha|$ drops from $\sim 0.5\%$ /month under CAPM to $\sim 0.1\%$ /month under FF3
- ▶ R^2 rises from ~ 0.7 to ~ 0.9
- ▶ The size and value loadings are statistically significant for most portfolios

Bottom line

FF3 is not perfect, but it is a much better description of the cross-section of expected returns than CAPM.

Quick check

A mutual fund has run for 10 years. Regressing its excess returns on the three Fama–French factors gives:

$$\hat{\alpha} = 0.20\%/month, \quad t(\alpha) = 2.4, \quad \hat{\beta}_m = 0.95, \quad \hat{\beta}_S = +0.6, \quad \hat{\beta}_H = -0.3$$

Which is the best one-sentence description of this fund?

- A Tracks the market closely, with no statistically significant risk-adjusted alpha.
- B A small-cap, growth-tilted fund with positive risk-adjusted alpha.
- C A large-cap value fund whose performance is fully explained by its factor exposures.

Beyond three factors

- ▶ **Momentum** (Carhart 1997): buy past winners, sell past losers
- ▶ **Profitability** and **Investment** (Fama–French 2015)

Fama–French 5-factor model

The modern standard.

Ken French's data library (factors, portfolios, breakpoints)

From FF3 to new return predictors

Key idea: Researchers use FF3 alpha to test whether a new characteristic matters beyond standard factor exposures.

Once we control for market, size, and value exposure:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{m,p}(R_{m,t} - R_{f,t}) + \beta_{S,p}R_{SMB,t} + \beta_{H,p}R_{HML,t} + \varepsilon_{p,t}$$

- ▶ If $\alpha_p \approx 0$, FF3 explains the strategy's average return.
- ▶ If $\alpha_p > 0$ and statistically significant, the strategy earns returns not explained by FF3.
- ▶ This is why researchers test governance, pollution, and other characteristics.

Governance, agency risk, and stock returns

Gompers, Ishii & Metrick (2003), Quarterly Journal of Economics

Strategy

Build a **Governance Index** from 24 shareholder-rights provisions. Long firms with *strong* shareholder rights; short firms with *weak* shareholder rights.

Sign of FF α

Positive — the strategy earns a significant positive FF α .

Proposed risk mechanism

Agency risk: weak governance exposes investors to entrenchment, inefficient investment, and value destruction.

CORPORATE GOVERNANCE AND EQUITY PRICES*

PAUL GOMPERS

JOY ISHII

ANDREW METRICK

Shareholder rights vary across firms. Using the incidence of 24 governance rules, we construct a “Governance Index” to proxy for the level of shareholder rights at about 1500 large firms during the 1990s. An investment strategy that bought firms in the lowest decile of the index (strongest rights) and sold firms in the highest decile of the index (weakest rights) would have earned abnormal returns of 8.5 percent per year during the sample period. We find that firms with stronger shareholder rights had higher firm value, higher profits, higher sales growth, lower capital expenditures, and made fewer corporate acquisitions.

TABLE VI
PERFORMANCE-ATTRIBUTION REGRESSIONS FOR DECILE PORTFOLIOS

	α	<i>RMRF</i>	<i>SMB</i>	<i>HML</i>	<i>Momentum</i>
Democracy-Dictatorship	0.71** (0.26)	-0.04 (0.07)	-0.22* (0.09)	-0.55* (0.10)	-0.01 (0.07)
$G \leq 5$ (Democracy)	0.29* (0.13)	0.98** (0.04)	-0.24** (0.05)	-0.21** (0.05)	-0.05 (0.03)
$G = 6$	0.22 (0.18)	0.99** (0.05)	-0.18** (0.06)	0.05 (0.07)	-0.08 (0.04)
$G = 7$	0.24 (0.19)	1.05** (0.05)	-0.10 (0.07)	-0.14 (0.08)	0.15** (0.05)
$G = 8$	0.08 (0.14)	1.02** (0.04)	-0.04 (0.05)	-0.08 (0.06)	0.01 (0.04)
$G = 9$	-0.02 (0.12)	0.97** (0.03)	-0.20** (0.04)	0.14** (0.05)	-0.01 (0.03)
$G = 10$	0.03 (0.11)	0.95** (0.03)	-0.17** (0.04)	-0.00 (0.04)	-0.08** (0.03)
$G = 11$	0.18 (0.16)	0.99** (0.05)	-0.14* (0.05)	-0.06 (0.06)	-0.01 (0.04)
$G = 12$	-0.25 (0.14)	1.00** (0.04)	-0.11* (0.05)	0.16** (0.06)	0.02 (0.04)
$G = 13$	-0.01 (0.14)	1.03** (0.04)	-0.21** (0.05)	0.14* (0.06)	-0.08* (0.04)
$G \geq 14$ (Dictatorship)	-0.42* (0.19)	1.03** (0.05)	-0.02 (0.06)	0.34** (0.07)	-0.05 (0.05)

We estimate four-factor regressions (equation (1) from the text) of value-weighted monthly returns for portfolios of firms sorted by G . The calculation of G is described in Section II. The first row contains the results when we use the portfolio that buys the Democracy Portfolio ($G \leq 5$) and sells short the Dictatorship Portfolio ($G \geq 14$). The portfolios are reset in September 1990, July 1993, July 1995, and February 1998, which are the months after new data on G became available. The explanatory variables are *RMRF*, *SMB*, *HML*, and *Momentum*. These variables are the returns to zero-investment portfolios designed to capture market, size, book-to-market, and momentum effects, respectively. (Consult Fama and French [1993] and Carhart [1997] on the construction of these factors.) The sample period is from September 1990 through December 1999. Standard errors are reported in parentheses and significance at the 5 percent and 1 percent levels is indicated by * and **, respectively.

The pollution premium

Hsu, Li & Tsou (2023), Journal of Finance

Strategy

Measure firm-level **toxic-emissions intensity**. Long high-pollution firms; short low-pollution firms.

Sign of FF α

Positive — high-pollution firms earn higher returns even after FF adjustment.

Proposed risk mechanism

Environmental risk: exposure to lawsuits, production disruptions, and tightening regulation; not captured by the three standard factors.

The implementation

1. **Get returns** for the portfolio (or stock) you want to evaluate
2. **Download the three factors** (and the risk-free rate) from Ken French's data library
3. **Merge** the two by date
4. **Compute excess returns** of the portfolio: $R_p - R_f$
5. **Run the regression** in R, Python, or Stata
6. **Read** $\hat{\alpha}$, $\hat{\beta}$ s, and their t-stats

A detailed walk-through is in the *Implementation Handout* (separate PDF).

Readings

Start here:

- ▶ Fama & French (1992), “The cross-section of expected stock returns,” *Journal of Finance*, 47, 427–465. Read the introduction and Section I.

If you want more depth:

- ▶ Fama & French (1993), “Common risk factors in the returns on stocks and bonds,” *Journal of Financial Economics*, 33, 3–56. The full construction is in Sections 2 and 3.

Applications:

- ▶ Gompers, Ishii & Metrick (2003), “Corporate governance and equity prices,” *Quarterly Journal of Economics*, 118, 107–155.
- ▶ Hsu, Li & Tsou (2023), “The pollution premium,” *Journal of Finance*.